A Citation Index for Federal Regulations

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Abstract

Federal regulations are required to rest on reasoned decision-making and the best available science, yet no systematic database exists of the research actually cited in the rulemaking process. We construct the first comprehensive citation index for federal regulations by extracting and cataloging every academic citation from 113,712 final rules published in the Federal Register between 1995 and 2025. Using large language models to parse regulatory text and CrossRef to enrich bibliographic metadata, we identify 85,327 citation instances spanning 54,912 unique papers by 66,178 unique authors. We find that citations to academic research have grown substantially—from roughly 1,100 per year in the late 1990s to nearly 9,000 in 2024—and vary widely across agencies in volume and disciplinary focus. Health and Human Services and the EPA account for nearly 40% of all citations. Where possible, we resolve working papers to their final published versions, substantially improving journal attribution. This dataset enables new research on the role of evidence in policymaking.

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1 Introduction

The Administrative Procedure Act (APA) requires federal agencies to engage in "reasoned decision-making" when promulgating regulations. Executive orders dating back decades—from President Reagan's Executive Order 12291 to President Clinton's E.O. 12866 and its successors—mandate cost-benefit analysis grounded in evidence. The Supreme Court's 2024 decision in *Loper Bright Enterprises v. Raimondo*, which overturned *Chevron* deference, further heightened the importance of the evidentiary record: courts now exercise independent judgment over whether agency interpretations are consistent with statutory mandates, making the quality and breadth of cited evidence a potential factor in judicial review.

Despite the centrality of evidence to regulatory legitimacy, no systematic database exists of the research actually cited in federal regulations. Scholars studying the regulatory process have examined rulemaking procedures, public comments, and cost-benefit analyses, but the specific academic papers, journals, and researchers that inform regulatory decisions have remained uncataloged. This gap limits our understanding of which disciplines and scholars influence policy, whether agencies draw on the best available science, and how the use of evidence has evolved over time.

This paper fills that gap. We construct the first comprehensive citation index for federal regulations by extracting every academic citation from the full text of 113,712 final rules published in the Federal Register from 1995 through early 2025. Our dataset identifies 85,327 citation instances, which we resolve to 54,912 unique papers by 66,178 unique authors. Of the roughly 113,000 final rules in our sample, 7,002 (about 6%) contain at least one academic citation.

The resulting dataset functions like a Google Scholar for federal regulations: for each paper, we record how many times it was cited in regulations, by which agencies, and in which years. For each author, we aggregate total regulatory citations across all their work. For each agency, we can identify the journals and researchers most frequently relied upon.

Three broad findings emerge. First, citations to academic research in federal regulations have grown substantially over the past three decades, from roughly 1,100 citations per year in the late 1990s to nearly 9,000 in 2024. Second, citation patterns vary dramatically across agencies, reflecting disciplinary specialization: the Department of Health and Human Services (HHS) cites medical journals such as the *New England Journal of Medicine* and *JAMA*, the Environmental Protection Agency (EPA) relies on environmental health and atmospheric science research, and the Securities and Exchange Commission (SEC) draws heavily on finance and economics journals. Third, the evidence base is broad: our index spans over 48,000 distinct journal titles, and the most-cited single journal accounts for fewer than 1% of all citations.

2 Data and Methods

2.1 Corpus

Our source corpus comprises the full text of every final rule published in the Federal Register from 1995 through January 2025—113,712 documents in total. We obtained document metadata (title, publication date, issuing agencies, document number) from the Federal Register API and downloaded the full text of each rule.

2.2 Citation Extraction

We extract academic citations in two stages. First, we identify footnotes in each document using the numbered-footnote convention of Federal Register typesetting. We pre-filter footnotes to remove legal citations (citations to the Code of Federal Regulations, United States Code, court cases, and other legal authorities) using regular expressions, retaining only footnotes that potentially contain references to academic or scientific literature.

Second, we use a large language model (Claude, Anthropic) to parse the remaining footnotes and extract structured citation information: author names, title, journal or publisher, year of publication, and DOI when available. To handle documents with large numbers of footnotes, we chunk footnotes into batches of 50 and process each batch independently. We use the Anthropic Message Batches API, processing the full corpus across nine batches organized by publication year.

2.3 Enrichment and Disambiguation

We enrich extracted citations by querying the CrossRef API, matching on title and author to retrieve DOIs, canonical journal names, ISSNs, and publisher information. Approximately 36% of citation occurrences (and 33% of unique papers) receive a DOI through this process.

Journal names are standardized using a multi-source registry built from the Web of Science citation indices (SCIE, ESCI, AHCI), Journal Citation Reports abbreviation tables, and manual correction files. This registry maps 48,757 canonical journal titles with 37,276 known ISSNs.

Author names undergo a disambiguation pipeline that normalizes name variants, resolves bare last names (common in parenthetical citations such as "Smith et al. (2007)") using CrossRef structured data where available, and applies a Union-Find algorithm with corroborating evidence—including co-authorship overlap, shared papers, and CrossRef identity matching—to merge 33,541 name variants into 66,178 unique author identities.

Paper deduplication proceeds in two passes: exact grouping by DOI and normalized title–year combinations, followed by fuzzy title matching. An iterative enrichment step propagates metadata across matching citations within the corpus.

Finally, we resolve working papers to their published versions. CrossRef title matching sometimes attributes citations to SSRN (the Social Science Research Network) when someone has uploaded a copy there, even if the original citation references a government report, law review article, or other source. We address this in two steps. First, we query the OpenAlex API by title for each SSRN-matched paper, validate matches using author-name overlap, and replace the SSRN attribution with the published journal when a match is found. Of 1,075 unique SSRN papers in our corpus, we resolve 500 (47%) to published journal articles, with top destinations including the *Journal of Financial Economics*, *The Journal of Finance*, and the *Review of Financial Studies*. Second, for the remaining 575 papers where no published version is found, we revert to the original citation information extracted from the regulatory text—since SSRN is not the true source of these citations, which are typically government reports, policy documents, or working papers. We apply the same reversion logic to other database platforms that CrossRef sometimes matches to incorrectly (e.g., Choice Reviews Online, PsycEXTRA Dataset, CFA Digest).

3 Results

3.1 Summary Statistics

Table 1 presents an overview of the dataset. From 113,712 final rules, we extract 85,327 citation instances from 7,002 regulations (6.2%). These resolve to 54,912 unique papers by 66,178 unique authors. Reports and gray literature constitute the largest share of citations (38.5%), followed by unclassified references (29.8%) and peer-reviewed journal articles (25.4%).

3.2 Citations Over Time

Figure 1 displays the number of academic citations in final rules by regulation year from 1996 to 2024. Citations have grown roughly eightfold, from approximately 1,100 in 1996 to 8,778 in 2024. Growth has been uneven, with notable spikes in years when major regulations were finalized—for example, 2016 saw 6,289 citations, driven in part by large EPA and HHS rulemakings. The overall trend reflects both an increase in the number of evidence-intensive regulations and a deepening of the evidentiary record within individual rules.

Table 1: Summary Statistics

Measure	**Count**
Final rules in corpus (1995–2025)	113,712
Rules with ≥ 1 academic citation	7,002 (6.2%)
Total citation instances	85,327
Unique papers (after deduplication)	54,912
Unique authors (after disambiguation)	66,178
Unique journal titles	48,757
Citations with DOI or CrossRef match	30,869 (36.2%)
By citation type:	
Reports & gray literature	32,822 (38.5%)
Other / unclassified	25,413 (29.8%)
Journal articles	21,683 (25.4%)
Books	2,938 (3.4%)
Working papers	1,367 (1.6%)
Conference papers	1,040 (1.2%)

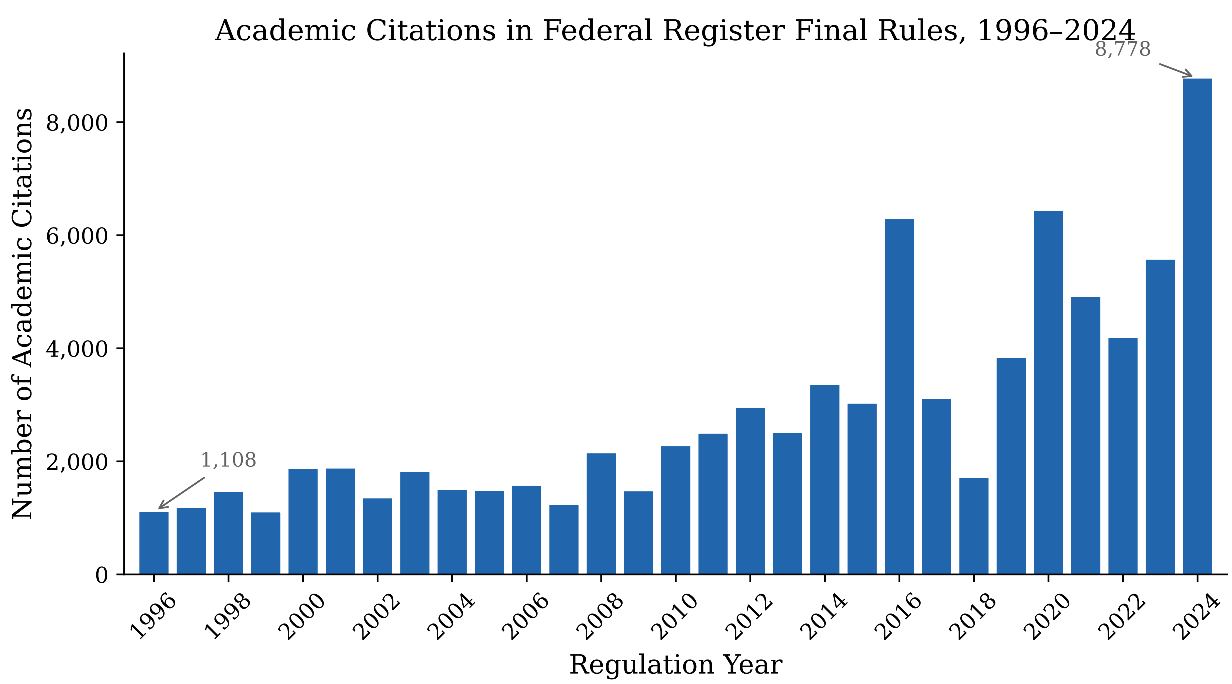

Figure 1: Academic citations in Federal Register final rules by regulation year, 1996–2024.

3.3 Citations by Agency

Citation volume varies dramatically across agencies. Table 2 reports the top 25 agencies by total citation count (excluding sub-agencies that overlap with parent department totals). HHS leads with 17,093 citations, followed by the EPA (15,743) and the Department of the Interior (12,142). Together, these three departments account for over half of all citations.

Table 2: Top 25 Agencies by Total Academic Citations, 1995–2025

Rank	Agency	Citations
1	Health and Human Services Department	17,093
2	Environmental Protection Agency	15,743
3	Interior Department	12,142
4	Commerce Department	9,164
5	Labor Department	6,350
6	Securities and Exchange Commission	5,668
7	Transportation Department	4,739
8	Energy Department	3,487
9	Treasury Department	2,857
10	Agriculture Department	2,593
11	Consumer Financial Protection Bureau	2,118
12	Homeland Security Department	1,763
13	Education Department	1,594
14	Internal Revenue Service	1,378
15	Commodity Futures Trading Commission	1,012
16	Federal Communications Commission	927
17	Federal Energy Regulatory Commission	882
18	Federal Reserve System	788
19	Justice Department	761
20	Agricultural Marketing Service	669
21	Federal Trade Commission	666
22	Housing and Urban Development Department	581
23	Federal Deposit Insurance Corporation	543
24	Comptroller of the Currency	497
25	Veterans Affairs Department	233

3.4 Top Journals

Table 3 lists the 25 most-cited journals. Medical journals dominate the top of the list, driven by HHS rulemaking: the *New England Journal of Medicine* ranks first, followed by *Health*

Affairs and *JAMA*. Finance and economics journals—the *Journal of Financial Economics* and *The Journal of Finance*—rank fourth and fifth after resolving SSRN working papers to their published venues. Environmental journals reflect EPA's evidence base. The full top-100 list appears in Appendix Table 6.

Table 3: Top 25 Most-Cited Journals

Rank	**Journal**	**Citations**
1	New England Journal of Medicine	427
2	Journal of the American Medical Association	412
3	Health Affairs	389
4	Journal of Financial Economics	323
5	The Journal of Finance	289
6	Environmental Health Perspectives	273
7	MMWR. Morbidity and Mortality Weekly Report	211
8	Environmental Science & Technology	197
9	SAE Technical Paper Series	195
10	Annals of Internal Medicine	173
11	Journal of the American Geriatrics Society	170
12	American Economic Review	166
13	Energy Policy	141
14	Journal of General Internal Medicine	140
15	Atmospheric Environment	136
16	Medical Care	136
17	Quarterly Journal of Economics	128
18	JAMA Network Open	125
19	Review of Financial Studies	125
20	The Lancet	119
21	Proc. Natl. Acad. Sci. USA	114
22	Health Services Research	113
23	Circulation: Cardiovascular Quality and Outcomes	108
24	British Medical Journal	107
25	American Journal of Kidney Diseases	106

3.5 Disciplinary Specialization Across Agencies

The citation patterns reveal strong disciplinary specialization. Table 4 shows the top five journals cited by major regulatory agencies. HHS relies heavily on clinical medicine journals. The EPA draws on environmental health and atmospheric science. The SEC's top journals—after resolving SSRN working papers to their published venues—are the leading finance and accounting journals. The Department of Labor's citations span health economics and labor economics. Interior and Commerce cite ecology and wildlife journals, reflecting Fish and Wildlife Service and NOAA rulemaking.

Table 4: Top 5 Journals by Agency

Agency	**Journal**	**Cites**	**Agency**	**Journal**	**Cites**
	N. Engl. J. Med.	383		J. Financial Economics	286
	Health Affairs	355		J. Finance	221
HHS	JAMA	333	*SEC*	J. Accounting & Econ.	88
	J. Am. Geriatrics Soc.	170		Accounting Review	79
	Annals of Internal Med.	158		Rev. Financial Studies	72
	Environ. Health Persp.	257		Health Affairs	69
	Environ. Sci. & Tech.	180		Am. Economic Review	44
EPA	Atmospheric Environ.	133	*Labor*	Q. J. Economics	39
	SAE Technical Papers	107		N. Engl. J. Med.	39
	J. Air & Waste Mgmt.	72		MMWR	27
	J. Wildlife Mgmt.	26		Marine Mammal Sci.	21
	J. Wildlife Diseases	10		Endangered Species Res.	11
Interior	JAWRA	7	*Commerce*	Fisheries Research	10
	Freshwater Biology	7		Copeia	7
	Poultry Science	7		Pathology	6
	SAE Technical Papers	136		Energy Policy	90
	Environ. Health Persp.	37		Bull. Exp. Biol. Med.	64
Transportation	Energy Policy	33	*Energy*	Rev. Economic Studies	30
	Transp. Research Record	33		J. Illuminating Eng.	23
	Accident Anal. & Prev.	28		J. Geophys. Res.: Atmos.	22
	Health Affairs	65		J. Food Protection	51
	N. Engl. J. Med.	25		Am. J. Agricultural Econ.	28
Treasury	Am. Economic Review	22	*Agriculture*	Am. J. Clinical Nutrition	23
	Finance & Econ. Disc. Ser.	21		Poultry Science	22
	Forefront Group	19		Agricultural Statistics	21
	J. Economic Perspectives	22		Finance & Econ. Disc. Ser.	17
	Violence Against Women	19		Federal Reserve Bulletin	17
Education	Econ. of Education Rev.	18	*CFPB*	Q. J. Economics	16
	Am. Economic Journal	16		Am. Economic Review	16
	Am. Economic Review	13		Supreme Ct. Econ. Rev.	16

3.6 Top-Cited Authors in HHS Regulations

Table 5 reports the 25 most-cited authors in HHS regulations. The list is dominated by health services and outcomes researchers whose work directly informs Medicare quality measurement, hospital readmission policy, and payment reform. Karen Joynt (now Joynt Maddox) of Washington University ranks first with 130 citations, primarily for studies on social risk factors and hospital performance. Harlan Krumholz (Yale) and Arnold Epstein (Harvard) follow, with work on hospital quality measurement and Medicaid. The concentration of citations among researchers studying Medicare quality and hospital readmissions reflects CMS's intensive use of evidence in designing value-based purchasing and readmission reduction programs.

Table 5: Top 25 Most-Cited Authors in HHS Regulations

	Author	**Cites**	**Primary Research Area**
1	Karen E. Joynt	130	Hospital quality, social risk (Wash. U.)
2	Harlan M. Krumholz	115	Cardiovascular outcomes, quality (Yale)
3	Arnold M. Epstein	105	Health policy, Medicaid (Harvard)
4	Yao-Zu Wang	91	Hospital quality measurement
5	Marc N. Elliott	82	Patient experience surveys (RAND)
6	Rachael Zuckerman	79	Social risk factors, Medicare
7	Alan M. Zaslavsky	58	Health disparities measurement (Harvard)
8	Vincent Mor	52	Nursing home quality (Brown)
9	Sharon-Lise T. Normand	52	Statistical methods, quality (Harvard)
10	E. John Orav	51	Biostatistics, outcomes (Harvard)
11	Susannah M. Bernheim	49	Hospital readmissions (Yale/CMS)
12	David W. Bates	49	Patient safety, health IT (Harvard)
13	Peter K. Lindenauer	47	Hospital quality, readmissions
14	Ashish K. Jha	47	Health systems, quality (Brown)
15	Bruce E. Landon	46	Primary care, disparities (Harvard)
16	Kenneth J. Ottenbacher	45	Rehabilitation outcomes (UTMB)
17	Eric A. Coleman	45	Care transitions (U. Colorado)
18	David C. Grabowski	45	Long-term care (Harvard)
19	Stephen F. Jencks	44	Hospital readmissions (CMS)
20	Joseph P. Newhouse	43	Health economics (Harvard)
21	Zhenqiu Lin	43	Biostatistics, quality metrics (Yale)
22	Mark W. Williams	43	Readmissions, quality (U. Kentucky)
23	Amal N. Trivedi	43	Health equity, VA quality (Brown)
24	Elizabeth E. Drye	42	Quality measurement (Yale/CMS)
25	Lisa G. Suter	42	Surgical outcomes, readmissions (Yale)

4 Discussion

Several patterns in the data merit discussion. First, the dominance of reports and gray literature (38.5% of citations) alongside peer-reviewed journal articles (25.4%) reflects the practical nature of regulatory evidence. Agencies frequently cite government reports, technical documents, and industry studies that provide the applied data needed for cost-benefit analysis, risk assessment, and compliance cost estimation. These sources complement the peer-reviewed literature by providing the specific empirical inputs—such as dose-response functions, market data, or compliance cost surveys—that regulations require.

Second, the disciplinary composition of citations tracks closely with each agency's statutory mandate. HHS regulations, particularly those from CMS governing Medicare payment and quality, cite clinical outcomes research. EPA regulations cite environmental epidemiology and toxicology. SEC regulations cite financial economics—and notably, nearly half of the SEC's SSRN citations resolve to top finance journals (*JFE*, *Journal of Finance*, *Review of Financial Studies*), suggesting that regulators are citing cutting-edge research before it appears in final published form. This specialization suggests that regulators are drawing on domain-appropriate research rather than a generic evidence base.

Third, the growth in citations over time likely reflects multiple forces: increased statutory and executive-order requirements for evidence-based rulemaking, the growing availability of relevant research, and expanding regulatory ambition in areas such as healthcare, environmental protection, and financial regulation. The sharp increase in 2024 (8,778 citations) warrants further investigation to determine whether it reflects a structural shift or is driven by a small number of unusually citation-heavy rules.

Finally, the concentration of citations among a relatively small number of researchers raises questions about researcher influence on policy. The most-cited authors at HHS—health services researchers studying Medicare quality—have a direct pipeline from their published findings to regulatory provisions. Understanding this relationship between academic research and regulatory outcomes is an important avenue for future work.

5 Conclusion

We present the first comprehensive citation index for federal regulations, covering three decades of rulemaking and identifying over 85,000 citation instances across 54,912 unique papers. The dataset reveals substantial growth in the use of academic evidence in rulemaking, strong disciplinary specialization across agencies, and a broad but concentrated evidence base.

This citation index opens several lines of inquiry. Researchers can study whether regulations that cite more or higher-quality evidence are more likely to survive judicial review, particularly in the post-*Chevron* era. The dataset can inform debates about whether agencies rely on "best available science" or exhibit citation bias. And by identifying which researchers are most frequently cited in regulations, the index provides a new measure of policy impact that complements traditional academic citation counts.

The dataset and code are available at [repository URL].

A Top 100 Journals

Table 6: Top 100 Most-Cited Journals in Federal Regulations, 1995–2025

Rank	**Journal**	**Cites**	**Rank**	**Journal**	**Cites**
1	N. Engl. J. Med.	427	51	Management Science	67
2	JAMA	412	52	Nature	65
3	Health Affairs	389	53	Med. Care Research & Rev.	65
4	J. Financial Economics	323	54	J. Am. Coll. Cardiology	64
5	J. Finance	289	55	J. Banking & Finance	63
6	Environ. Health Persp.	273	56	J. Food Protection	62
7	MMWR	211	57	New York Times	62
8	Environ. Sci. & Tech.	197	58	J. Arthroplasty	61
9	SAE Technical Papers	195	59	J. Public Economics	61
10	Annals of Internal Med.	173	60	J. Hospital Medicine	59
11	J. Am. Geriatrics Soc.	170	61	Inhalation Toxicology	58
12	Am. Economic Review	166	62	Wall Street Journal	57
13	Energy Policy	141	63	Clinical Infectious Dis.	56
14	J. General Internal Med.	140	64	Infection Control & Hosp. Epi.	54
15	Atmospheric Environ.	136	65	Annals of Emergency Med.	54
16	Medical Care	136	66	Critical Care Medicine	52
17	Q. J. Economics	128	67	Am. J. Infection Control	51

Rank	Journal	Cites	Rank	Journal	Cites
18	JAMA Network Open	125	68	Transp. Research Record	50
19	Rev. Financial Studies	125	69	J. Policy Anal. & Mgmt.	49
20	Lancet	119	70	Yale Law Journal	47
21	PNAS	114	71	Am. J. Preventive Med.	46
22	Health Services Research	113	72	J. Health Economics	46
23	Circ.: Cardiovasc. Qual. Outcomes	108	73	Diabetes Care	44
24	BMJ	107	74	Clin. J. Am. Soc. Nephrology	44
25	Am. J. Kidney Diseases	106	75	Federal Reserve Bulletin	44
26	Science	105	76	medRxiv	44
27	Am. J. Public Health	104	77	CHEST	43
28	J. Political Economy	96	78	Environmental Pollution	42
29	Arch. Physical Med. & Rehab.	95	79	J. Am. Med. Directors Assoc.	42
30	Am. J. Respir. Crit. Care Med.	91	80	Atmos. Chem. & Physics	41
31	J. Accounting & Econ.	89	81	Kidney International	41
32	Accounting Review	87	82	Am. J. Epidemiology	41
33	Arch. Internal Med.	86	83	Rev. Economic Studies	41
34	Finance & Econ. Disc. Ser.	83	84	Financial Analysts J.	40
35	Rev. Econ. & Statistics	82	85	Occupational & Environ. Med.	40
36	Blood	80	86	Environmental Research	39

Rank	Journal	Cites	Rank	Journal	Cites
37	PLoS ONE	80	87	J. Am. Soc. Nephrology	39
38	J. Fin. & Quant. Anal.	77	88	Am. J. Clinical Nutrition	39
39	J. Economic Perspectives	75	89	J. Vascular & Interv. Radiology	38
40	Pediatrics	75	90	Obstetrics & Gynecology	38
41	JAMA Internal Medicine	75	91	Psychiatric Services	38
42	Bull. Exp. Biol. & Med.	74	92	Emerging Infectious Dis.	37
43	J. Air & Waste Mgmt. Assoc.	74	93	Int. J. Environ. Res. Pub. Health	37
44	J. Accounting Research	72	94	Stroke	37
45	J. Clinical Oncology	71	95	Academic Medicine	36
46	Circulation	70	96	Am. J. Industrial Med.	36
47	Am. J. Industrial Med.	69	97	J. Geophys. Res.: Atmos.	36
48	J. Geophys. Res.: Atmos.	69	98	Lancet Infectious Diseases	35
49	Water, Air, & Soil Pollution	68	99	Clinical Chemistry	35
50	Nature	65	100	Nephrology Dial. Transplant.	35

B Data Construction Methodology

This appendix provides a detailed description of the data construction pipeline, from the raw Federal Register corpus through the final citation index. The pipeline consists of six stages: (1) corpus assembly, (2) citation extraction, (3) bibliographic enrichment, (4) journal standardization, (5) author disambiguation and paper deduplication, and (6) working paper resolution. Figure 2 provides a schematic overview.

Stage 1: Corpus Assembly
Federal Register API → 113,712 final rule texts (1995–2025)

Stage 2: Citation Extraction
Footnote parsing → Legal citation filtering → Chunking (batches of 50) → Claude Sonnet 4 (Batch API) → Structured JSON

Stage 3: Bibliographic Enrichment
CrossRef API (DOI lookup → title search → author+year fallback)

Stage 4: Journal Standardization
ISSN-based registry from Web of Science indices + JCR + manual fixes

Stage 5: Disambiguation and Deduplication
Paper dedup (DOI → title+year → fuzzy match) → Author disambiguation (Union-Find)

Stage 6: Working Paper Resolution
SSRN → OpenAlex API → published journal; database misattribution reversion

Figure 2: Data construction pipeline overview.

B.1 Corpus Assembly

We obtained metadata for every document published in the Federal Register from 1995 through January 2025 via the Federal Register API (`federalregister.gov/api`). For each document classified as a "Final Rule," we downloaded the full text, yielding 113,712 documents. Document metadata includes the title, publication date, document number, issuing agencies, and regulatory identification numbers.

The full-text files follow the typographic conventions of the Federal Register, with numbered footnotes marked by backslash-delimited superscripts (e.g., `\1\` through `\N\`). These footnotes contain the vast majority of citations in regulatory text, including references to academic literature, government reports, legal authorities, and other sources.

B.2 Citation Extraction

Citation extraction proceeds in three sub-steps: footnote identification, legal citation filtering, and large language model (LLM) classification.

B.2.1 Footnote Identification

We extract numbered footnotes from each document using the Federal Register's typographic convention. Each document's text is segmented at section dividers (lines of ten or more hyphens), and within sections beginning with a numbered footnote marker, we apply a regular expression to capture each footnote's number and content:

```
\\(\d+)\\\s+(.*?)(?=(?:\\\d+\\|\Z))
```

This pattern matches the backslash-delimited footnote number and captures all text up to the next footnote marker or end of section. Footnotes shorter than 20 characters and cross-references (e.g., those beginning with "Ibid.," "Id.," or "See supra") are excluded.

In addition to footnotes, we extract parenthetical citations of the form "Author et al. (Year)" or "(Author, Year)" from the body text, along with 300 characters of surrounding context. We also identify formal reference sections using header patterns (e.g., "References," "Bibliography," "Works Cited") and extract individual entries. These supplementary extractions capture citations that appear outside the footnote apparatus.

B.2.2 Legal Citation Pre-Filtering

Federal Register footnotes frequently cite legal authorities rather than academic sources. To reduce unnecessary API costs, we pre-filter footnotes locally using a battery of regular expressions that match common legal citation formats. Table 7 lists the principal patterns. Footnotes matching any of these patterns are excluded before submission to the language model. This pre-filtering step removes the majority of footnotes—which cite CFR sections, statutes, court opinions, and other legal materials—while retaining those that potentially reference academic or scientific literature.

Table 7: Legal Citation Patterns Used for Pre-Filtering

Citation Type	**Pattern (simplified)**
Code of Federal Regulations	`\d{1,2} CFR \d+`
United States Code	`\d{1,2} U.S.C. \d+`
Federal Reporter (2d, 3d)	`\d+ F.2d \d+`, `\d+ F.3d \d+`
U.S. Reports	`\d+ U.S. \d+`
Supreme Court Reporter	`\d+ S.Ct. \d+`
Public Law	`Pub. L. No. \d+`
Federal Register	`Fed. Reg.`
Statutes at Large	`\d+ Stat. \d+`
Executive Orders	`E.O. \d+`
Court cases	`v. [A-Z]`

B.2.3 LLM-Based Extraction

We use Anthropic's Claude Sonnet 4 (`claude-sonnet-4-20250514`) to classify and parse the remaining footnotes into structured citation records. The model receives a system prompt instructing it to exclude legal citations and extract only academic and scientific references, and a user prompt containing the footnote text. For each citation identified, the model returns a JSON object with the following fields: the original citation text, parsed title, author names, publication year, journal or publisher name, DOI (if present), citation type (journal article, book, report, working paper, conference paper, dissertation, or other), a flag for truncated author lists ("et al."), and any organizational author.

Chunking. Documents with many footnotes are split into consecutive chunks of 50 footnotes each. Each chunk is submitted as a separate API request with a custom identifier encoding both the document number and chunk index. Without chunking, we found that the model would sometimes stop early on documents with hundreds of footnotes, producing incomplete extractions. Setting the maximum output length to 16,384 tokens per chunk and limiting each chunk to 50 footnotes ensures complete processing. During post-processing, results from all chunks belonging to the same document are recombined.

Batch API. All requests are submitted via Anthropic's Message Batches API, which processes requests asynchronously at a 50% discount relative to synchronous API calls. The full corpus of 113,712 documents was processed in nine batches organized by publication-year ranges, with each batch typically completing within 24 hours. Total API cost for the full extraction was approximately $550.

B.3 Bibliographic Enrichment

We enrich extracted citations by querying the CrossRef API (`api.crossref.org`) to obtain DOIs, canonical journal names, ISSNs, volume and page information, and structured author data. We use the "polite pool" (providing a contact email in the `mailto` parameter) for faster rate limits.

B.3.1 Three-Tier Matching Strategy

For each citation, we attempt to match it to a CrossRef record using a cascading strategy:

1. **DOI lookup.** If the extracted citation includes a DOI, we query CrossRef directly by DOI. This is the most reliable matching method and yields an exact match (match score = 1.0).

2. **Title search.** If no DOI is available and the title is at least 10 characters long, we search CrossRef by title (`query.title` parameter, returning up to 3 results). We compare the extracted title against each returned title using the `SequenceMatcher` algorithm and accept the best match if its similarity ratio exceeds 0.70.

3. **Author+year search.** If neither DOI nor title matching succeeds, and both an author name and year are available, we search CrossRef by the first author's last

name (`query.author` parameter, returning up to 10 results). We accept a match if the publication year matches exactly and, when a title is available, the title similarity exceeds 0.50 (a lower threshold reflecting the less precise search).

All CrossRef responses are cached to a local JSON file, keyed by DOI or normalized title, to avoid redundant API calls across processing runs.

B.3.2 Field Resolution

For each citation, we maintain parallel fields preserving the original LLM extraction (prefixed `claude_`) and the CrossRef match (stored in `crossref_raw`). "Final" fields are resolved by preferring CrossRef data when a match exists and falling back to the LLM extraction otherwise. For example, `final_journal` takes the CrossRef container title if matched, and the Claude-extracted journal name otherwise. This dual-track approach allows us to assess enrichment quality and revert to original extractions when CrossRef matching proves unreliable.

B.4 Journal Standardization

Journal names in academic citations appear in many variant forms: full titles, standard abbreviations, non-standard abbreviations, and occasional misspellings. We standardize journal names using an ISSN-based registry constructed from multiple authoritative sources.

B.4.1 Registry Construction

The journal registry is built by ingesting the following sources in priority order:

1. **Web of Science citation indices.** We load journal lists from the Science Citation Index Expanded (SCIE), Emerging Sources Citation Index (ESCI), and Arts & Humanities Citation Index (AHCI). Each provides the canonical journal title, print and electronic ISSNs, and Web of Science subject categories.

2. **Journal Citation Reports (JCR).** The JCR file provides mappings from standard 20-character abbreviated titles to full journal titles, along with index membership flags (SCIE, SSCI, AHCI, ESCI). This adds approximately 21,800 abbreviation-to-title mappings.

3. **Manual correction files.** We maintain three supplementary mapping files:

   - `journal_fixes.json`: 3,614 manual mappings for common abbreviations and name variants not covered by JCR (e.g., "JAMA" → "Journal of the American Medical Association").
   - `journals_hash.json`: additional abbreviation mappings from prior research.
   - `entity_hash.json`: abbreviation expansions for both journals and organizations.

4. **CrossRef container data.** From the CrossRef responses in our enriched citations, we extract container-title and ISSN pairs, adding any ISSN–title associations not already in the registry.

The resulting registry maps 48,757 canonical journal titles with 37,276 known ISSNs.

B.4.2 Normalization Algorithm

For each citation's journal name, the lookup proceeds through five steps:

1. Exact match against the lowercase title index.
2. Match after removing a leading article ("The").
3. Match against a "cleaned" index that strips periods, hyphens, commas, ampersands, and whitespace.
4. Cleaned match after removing a leading article.
5. If a match is found and both "The X" and "X" variants exist, unify to the variant with a known ISSN.

B.5 Paper Deduplication

The same paper may be cited in multiple regulations with slightly different title spellings, author orderings, or metadata. We deduplicate papers using a two-pass approach.

B.5.1 Pass 1: Exact Key Grouping

Each citation is assigned a deduplication key using the following hierarchy:

1. **DOI** (most reliable): normalized to lowercase with the `https://doi.org/` prefix removed.

2. **ISSN + volume + first page**: for citations with CrossRef data that includes these fields, catching cases where titles differ slightly.

3. **Normalized title + year**: the title is lowercased, stripped of punctuation, and truncated to 120 characters; combined with the publication year.

4. **First author + year + journal**: used when the title is too short or missing.

5. **Text hash**: a hash of the original citation text as a last resort.

Citations sharing the same key are grouped together. For each group, we select the citation with the most non-empty fields as the base record, merge author lists across all citations in the group, and record all unique citing documents.

B.5.2 Pass 2: Fuzzy Title Matching

After exact grouping, we perform a second pass to merge groups with nearly identical titles. We block candidate pairs by shared publication year and first author last name (for computational tractability), then compare titles using the `SequenceMatcher` algorithm. Groups whose titles exceed a similarity threshold of 0.90 are merged.

B.5.3 Iterative Enrichment

Two additional passes propagate metadata across the corpus:

- **Within-corpus matching**: Papers lacking a DOI or CrossRef match are compared against a "confident pool" of papers that do have CrossRef data. Matches with title similarity above 0.85 (within the same year and first-title-word block) inherit the confident paper's DOI, journal, publisher, and author data.
- **Cross-citation propagation**: Papers sharing the same deduplication key have their metadata merged, with non-empty fields from the best-matched source taking precedence.

B.6 Author Disambiguation

Author names in regulatory citations appear in many forms: full names ("Harlan M. Krumholz"), abbreviated names ("H. M. Krumholz"), initials only ("H.M.K."), last name only ("Krumholz"), and occasionally inverted or misspelled variants. Our disambiguation pipeline resolves these variants into unique author identities using a Union-Find algorithm with multiple types of corroborating evidence.

B.6.1 Name Normalization

Before disambiguation, all author names are normalized:

- Names are converted to lowercase with standardized "Last, First" format.
- Periods are removed from initials ("h m" rather than "h. m.").
- Repeated name segments are detected and removed (e.g., "doty, michelle m doty" → "doty, michelle m").

- Inverted names are detected heuristically: if the putative last name is very short with no vowels and the putative first name is longer and resembles a real surname, the names are swapped.
- A dictionary of 110 manual corrections (`author_fixes.json`) handles known edge cases.
- A curated dictionary of 9,873 entries (`citation_authors_dict.json`) maps raw citation text to full author name lists for cases where the original text was ambiguous.

B.6.2 CrossRef Name Bank

We construct a "name bank" from the structured author fields in CrossRef responses across the entire corpus. For each CrossRef author record containing a family name and given name, we index the full given name by the tuple (last name, first initial). This allows us to resolve abbreviated names: for example, if CrossRef records consistently associate the key ("krumholz," "h") with the given name "Harlan M.," we can expand "Krumholz, H." to "Krumholz, Harlan M." A list of 436 government agency names is used to distinguish organizational authors from personal names.

B.6.3 Union-Find with Corroborating Evidence

Author name variants are grouped by (last name, first initial) into candidate clusters. Within each cluster, we apply a Union-Find algorithm that merges two variants only when supported by at least one type of corroborating evidence:

1. **Same paper**: Two name variants that appear as authors of the same paper (by deduplication key) are merged unconditionally, since they must refer to the same person.
2. **Co-author overlap with name subsumption**: Two variants that share at least one co-author across their respective papers, *and* where one name is a strict subsumption of the other (e.g., "J. Smith" subsumes "John A. Smith" if "j" is a prefix of "john"),

are merged. The subsumption requirement prevents merging "J. Smith" with "Jane Smith" based on co-authorship alone.

3. **CrossRef structured name match**: If the CrossRef name bank maps both variants to the same unique full name, they are merged.

4. **Unique first-name stem**: If all full-name variants within a (last name, initial) group share a single first-name stem (e.g., all "M. Elliott" variants resolve to "Marc"), then initial-only variants are merged with the full-name form.

After the main Union-Find pass, two post-processing passes handle special cases:

- **Compound last names**: Detects cases where a compound last name has been misparsed (e.g., "Riegg Cellini, Stephanie" vs. "Cellini, Stephanie Riegg") and merges the variants.

- **Flipped names**: Detects cases where the first and last names have been swapped (e.g., "Ashish K, Jha" instead of "Jha, Ashish K") and corrects the canonical form.

For each cluster produced by the Union-Find algorithm, we select the longest name variant as the canonical form. Bare last names (e.g., "Smith" with no first name) are merged with a full-name canonical only if exactly one canonical author shares a paper with the bare name.

This procedure merges 33,541 name variants into 66,178 unique author identities.

B.7 Working Paper Resolution and Misattribution Correction

CrossRef's title-matching algorithm sometimes attributes citations to database or repository entries rather than the true source. The most common case involves the Social Science Research Network (SSRN): when a working paper has been uploaded to SSRN, CrossRef may match the citation to the SSRN entry rather than the original source—which in regulatory text is often a government report, policy document, or law review article, not an academic working paper at all.

B.7.1 SSRN Resolution via OpenAlex

For the 1,075 unique papers in our corpus attributed to "SSRN Electronic Journal" by CrossRef, we query the OpenAlex API (`api.openalex.org`) by title to determine whether a published journal version exists. For each OpenAlex result, we verify that (a) the source is classified as a journal (not a repository), (b) the DOI does not contain the SSRN prefix (`10.2139`), and (c) at least one author last name overlaps between our citation and the OpenAlex record. Of 1,075 SSRN papers, 500 (47%) resolve to published journal articles. The top destination journals are the *Journal of Financial Economics* (55 papers), *The Journal of Finance* (48), and the *Review of Financial Studies* (21). For resolved papers, we update the journal name, DOI, ISSN, and publication year to reflect the published version.

B.7.2 Misattribution Reversion

For the remaining 575 SSRN papers where no published version is found, the SSRN attribution is almost certainly a CrossRef artifact—the original regulatory citation typically references a government report, working paper, or policy document that someone subsequently uploaded to SSRN. For these cases, we revert to the original citation information extracted by the language model (i.e., the `claude_journal` field), discarding the CrossRef attribution.

We apply the same reversion logic to other databases and platforms where CrossRef title matching produces systematic misattributions. Table 8 lists the affected sources and the number of citation instances reverted.

In each case, the reverted citations retain a flag (`crossref_misattributed` or `ssrn_misattributed`) and preserve the original CrossRef-attributed journal name, allowing future analysts to review these decisions. The reversion restores the citation to whatever source information the

Table 8: Database and Platform Misattributions Reverted to Original Extraction

CrossRef-Attributed Source	**Citations Reverted**
SSRN Electronic Journal (unresolved)	943
Choice Reviews Online	421
PsycEXTRA Dataset	390
CFA Digest	76
Fuel and Energy Abstracts	24
SpringerReference	17
PsycTESTS Dataset	12
Other databases and eBook platforms	62
Total	**1,945**

language model extracted from the regulatory text—typically a government agency report, technical document, or unaffiliated working paper.

B.8 Output Data Files

The final citation index consists of three CSV files:

- **Citations file** (85,327 rows): One row per citation occurrence, preserving every instance of a citation in a regulation. Each row records the citing regulation (document ID, agencies, publication date), the cited paper (title, authors, year, journal, DOI), the citation type, and CrossRef match metadata.
- **Papers file** (54,912 rows): One row per unique paper after deduplication. Each row records the paper's bibliographic information, the number of times it is cited in regulations, the set of citing regulations and agencies, and flags for Web of Science index membership.
- **Authors file** (66,178 rows): One row per unique author after disambiguation. Each row records the canonical name, all observed name variants, total papers and citations, the most-cited journal, the citing agencies, and top co-authors.